A solution of bounded dynamic range yields a linearly ordered colouring with logarithmically many colours

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Editorial scope. One original result of the pinned author manuscript, counted once: a linearly ordered colouring with logarithmically many colours from a solution of bounded dynamic range (author label lemma:unbal).

Allowed standard prerequisites (editorial).

- Homogeneous linear system over the reals. The coordinates of a solution are real numbers, with M the largest absolute value and m the smallest positive absolute value.. Their ratio controls the number of colours.
- Linearly ordered colouring. A colouring in which the smallest colour appearing once in an edge is unique.. It is what the construction produces.
- Logarithmic colour classes. The colours are assigned by comparing the absolute value of a coordinate against powers of two, with even colours used for positive coordinates and odd ones for negative coordinates.. It is the construction whose correctness has to be checked.

Lemma 1. *Suppose we have a solution, u, to the homogeneous system where M is the maximal value of the absolute value of a coordinate and $m > 0$ is the minimal absolute value. Then, we can LO colour with $2 + \log_2(M/m)$ colours.*

Proof. For notational convenience let us instead require that the minimal colour in each edge should be unique. We can simply reverse the order of the colours at the end. By scaling we can assume $M = 1$. We use even colours for positive coordinates and we give x the colour 2ℓ if v_x is at most $2^{-(2\ell-1)}$ and strictly larger than $2^{-(2\ell+1)}$. For negative coordinates we use $2\ell + 1$ as the colour if v_x is between $-2^{-2\ell}$ (inclusive) and $-2^{-(2\ell+2)}$ (non-inclusive). Let us verify that this gives a correct colouring.

Take an edge (x, y, z) and suppose both x and y get the same colour 2ℓ. Then by the linear equation of the edge $v_z < -2^{-2\ell}$ and thus z has a colour below 2ℓ. The case of two vertices of odd colour is similar and as the bound on the number of colours is immediate, the lemma follows. □